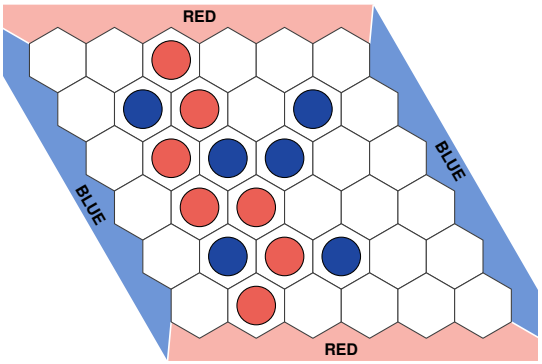
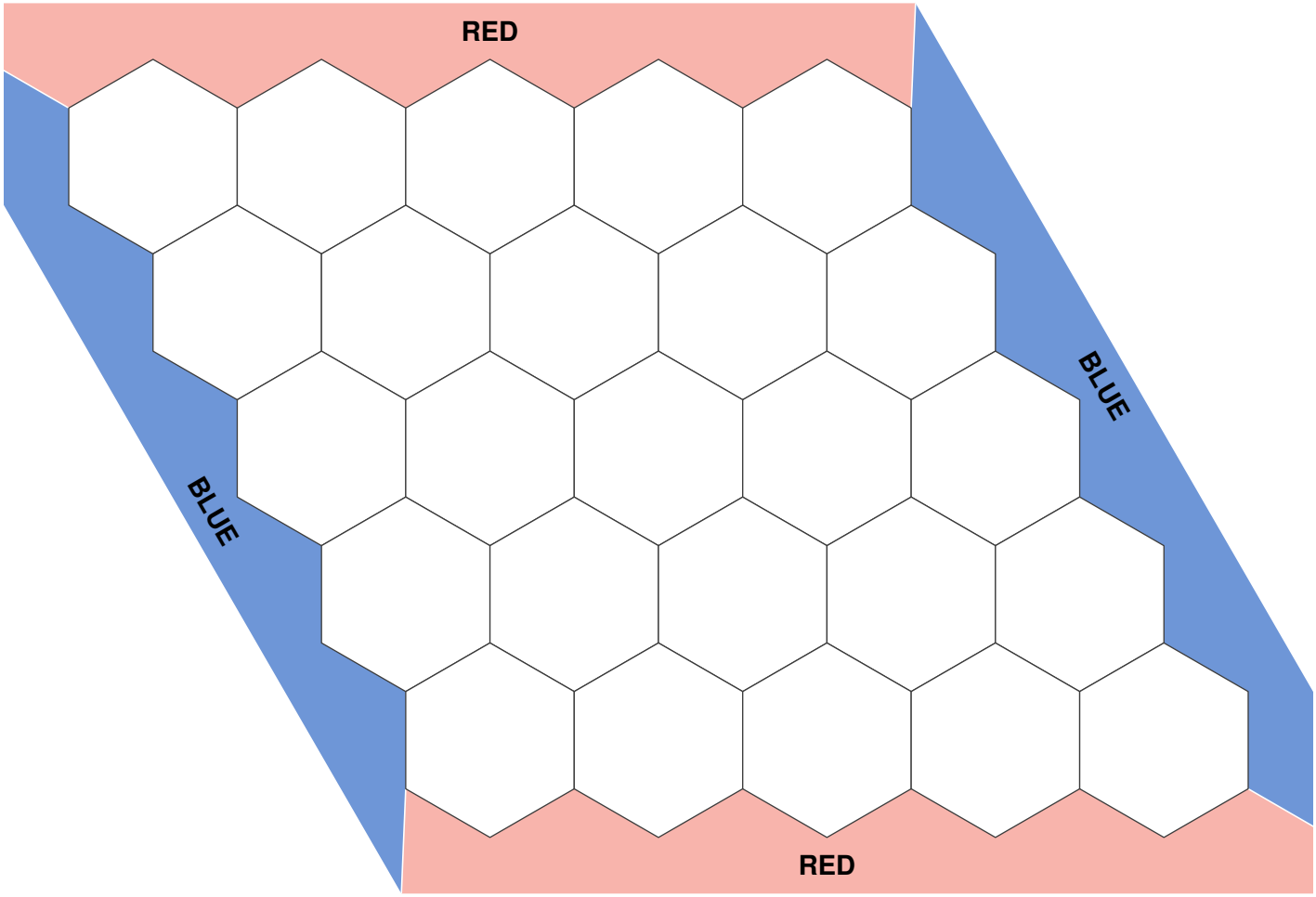
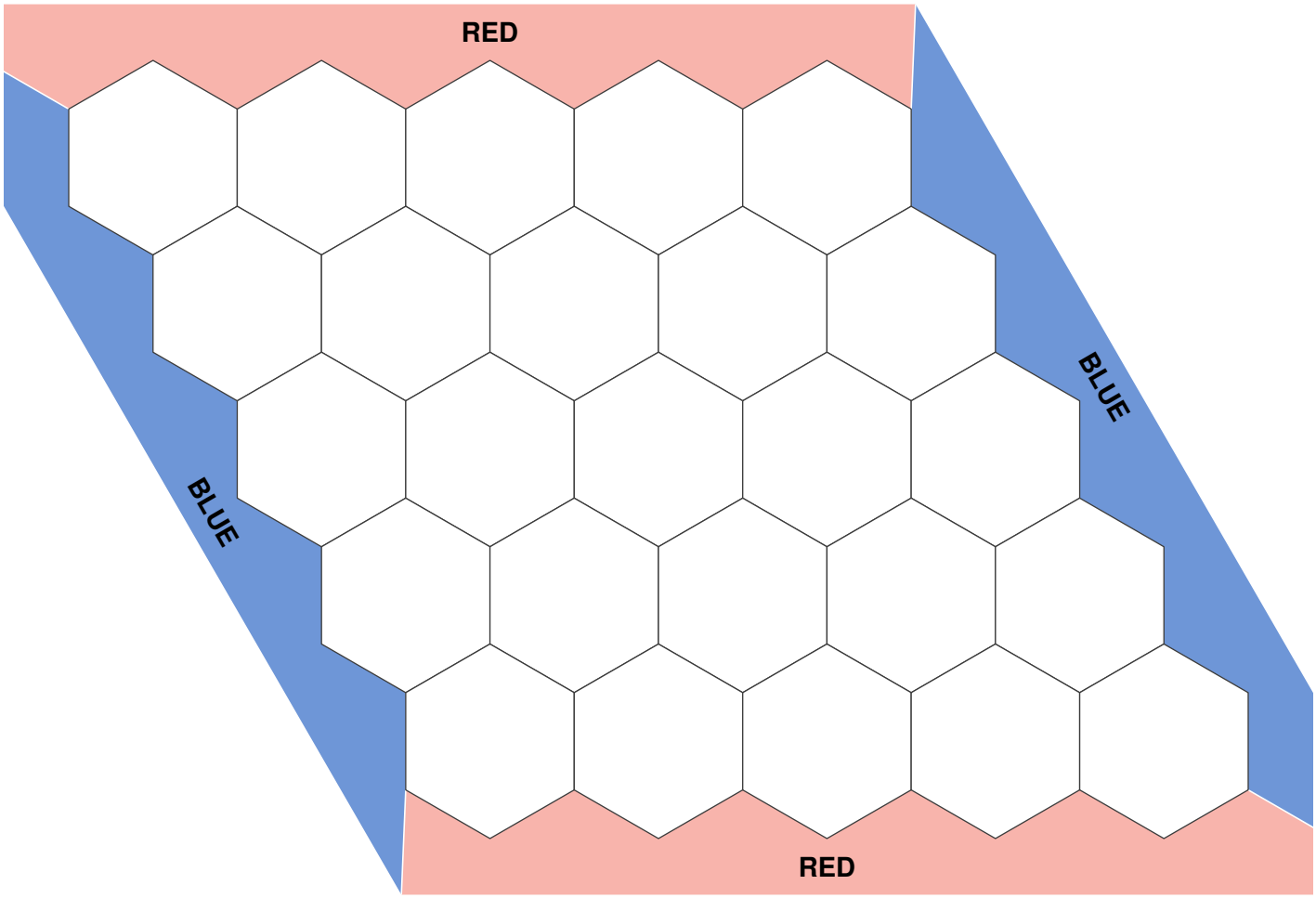
Red and Blue take turns putting a counter of their own color on an empty hexagon. Red wins by joining the two red edges with a chain of red hexagons. Blue wins by joining the two blue edges with a chain of blue hexagons. Each hexagon in a chain shares a side with the next one.



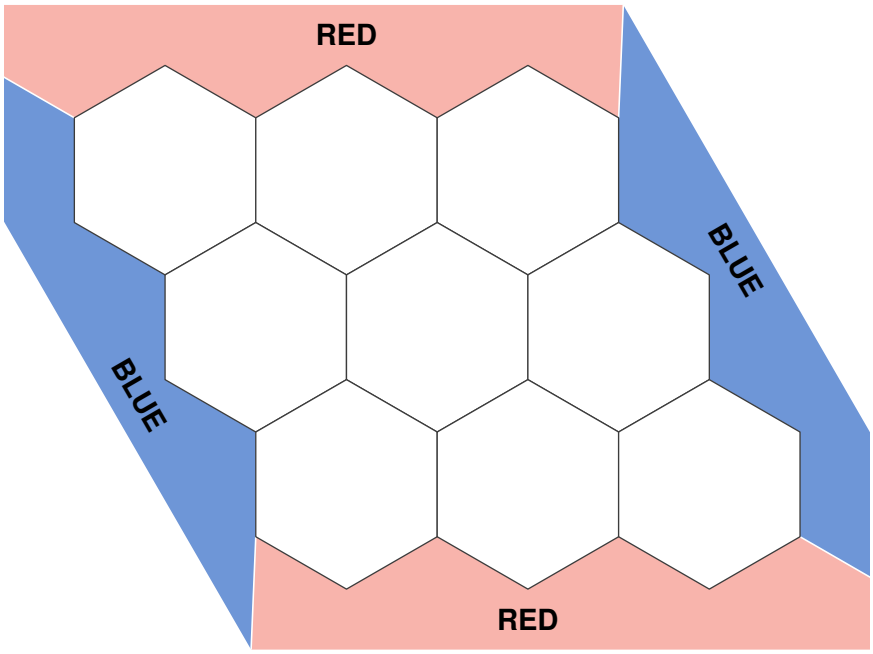
Problem 1: Play Hex on this board. Take turns going first.

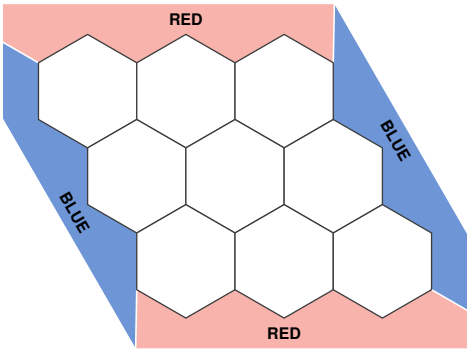


Problem 2: Can you put a red or a blue counter on every hexagon so that nobody wins? Can you do it so that Red and Blue both win?

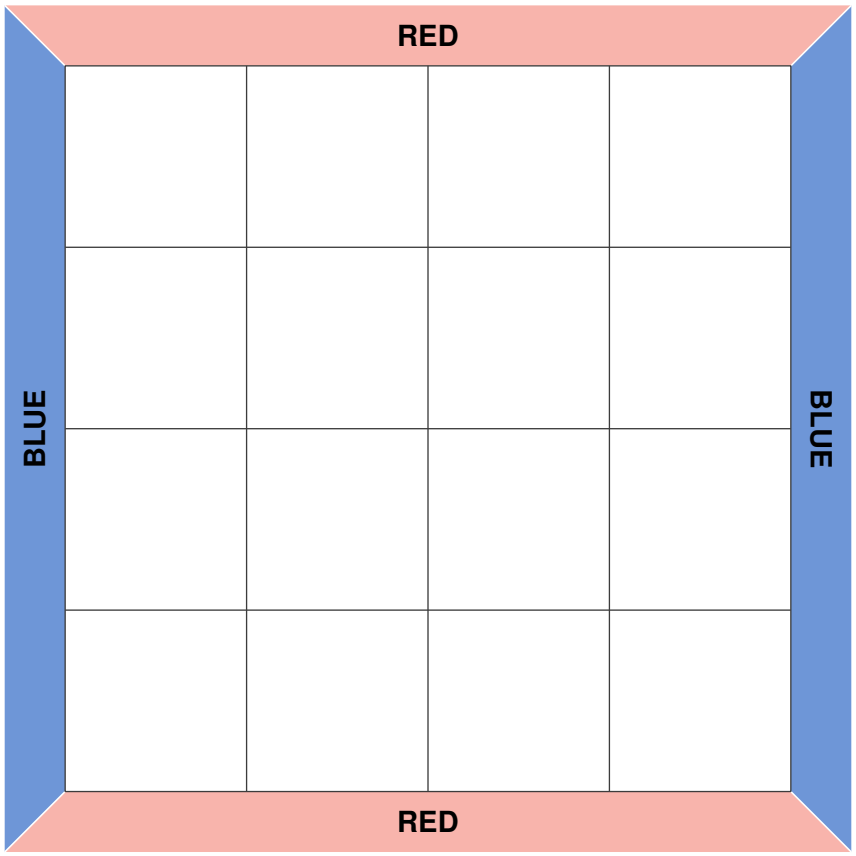


Problem 3: Red goes first on this board. Find every hexagon where Red can put the first counter and then be sure to win, however Blue plays. Put an X on those hexagons on the small board.



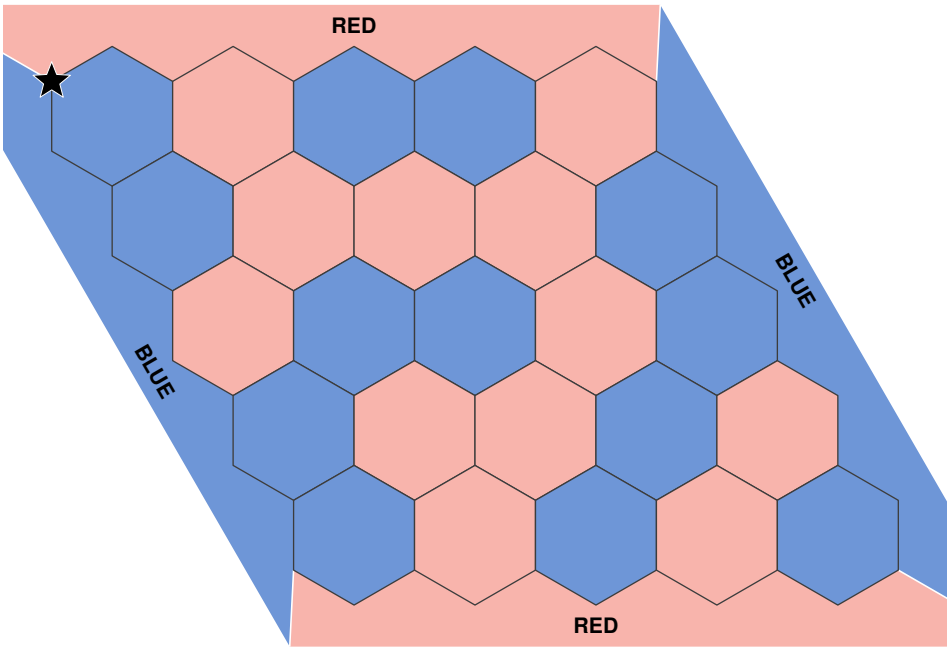


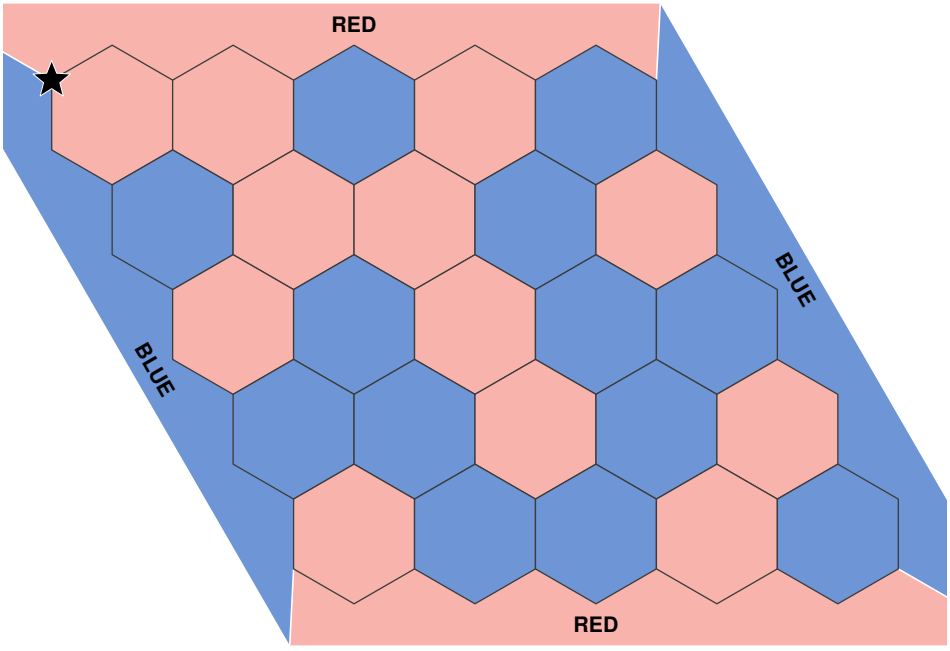
Problem 4: On this board of squares, a chain is made of squares that each share a side with the next. Put a red or a blue counter on every square so that nobody wins. Next, count squares that touch only at a corner as touching too. Fill the board so that Red and Blue both win.



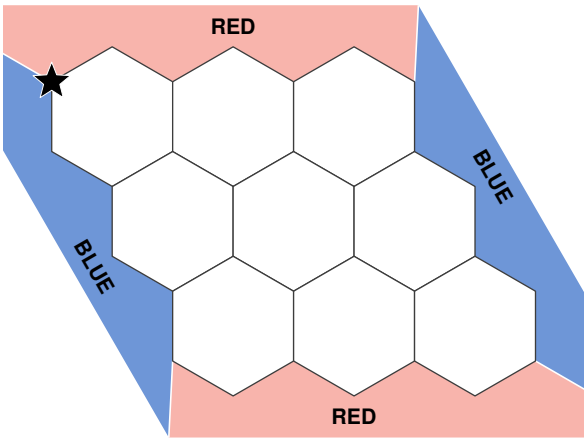
Problem 5: Explain why Red and Blue can never both have a winning chain on the same hexagon board.

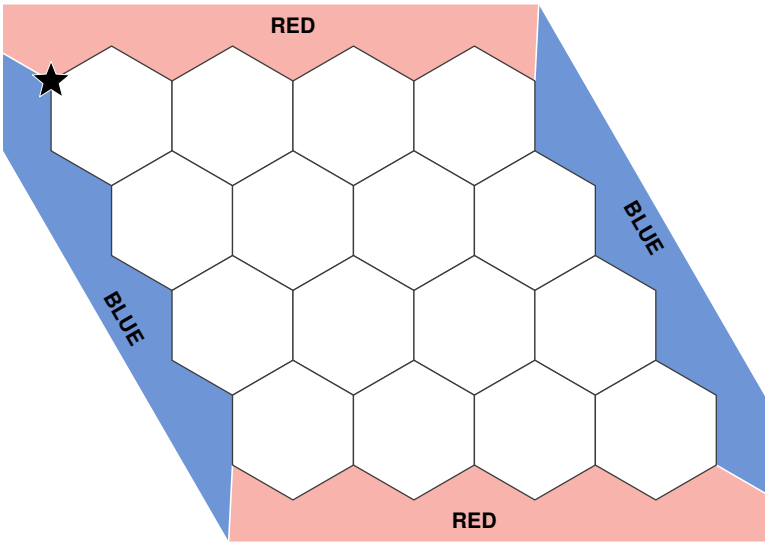
Problem 6: On each board, start at the star and draw a path along the sides of the hexagons, always with red on one side of the path and blue on the other. The colored edges count as red and blue. Follow the path as far as it goes. Where does it end?





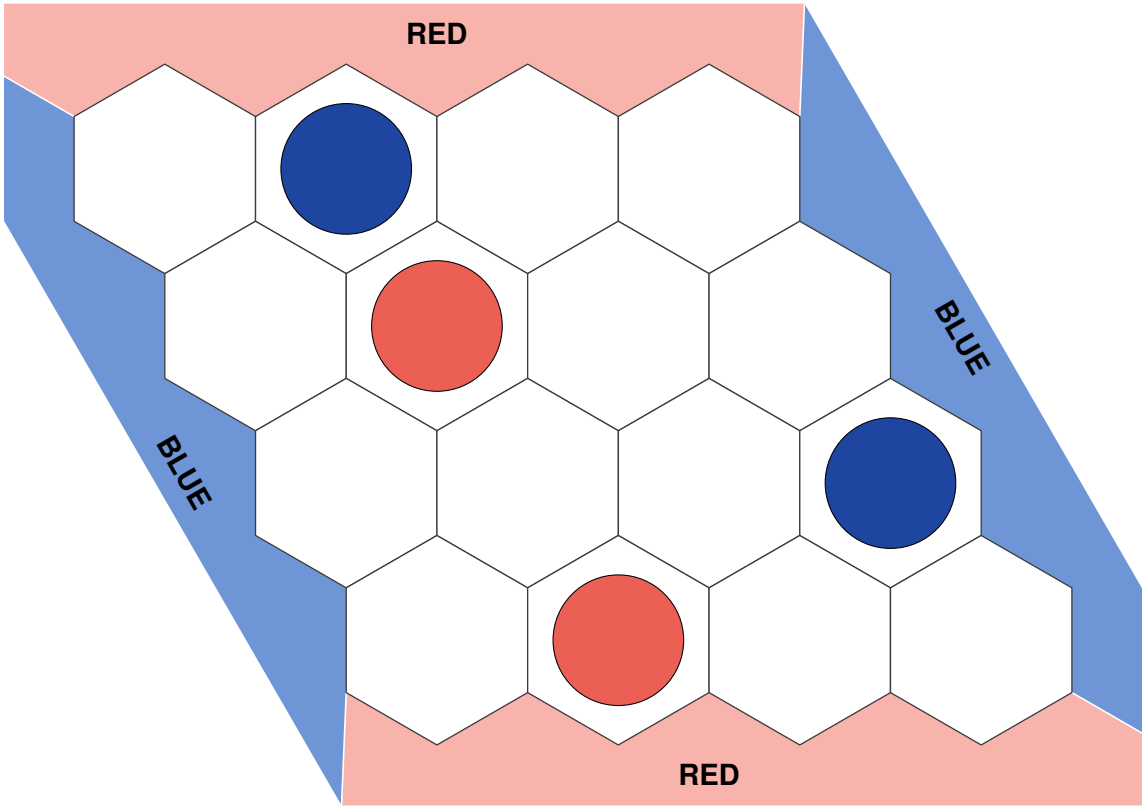
Problem 7: Color every hexagon on these boards red or blue, any way you like. Then draw the path from the star on each board, as in Problem 6. Can a path end anywhere other than the places where the paths in Problem 6 ended?

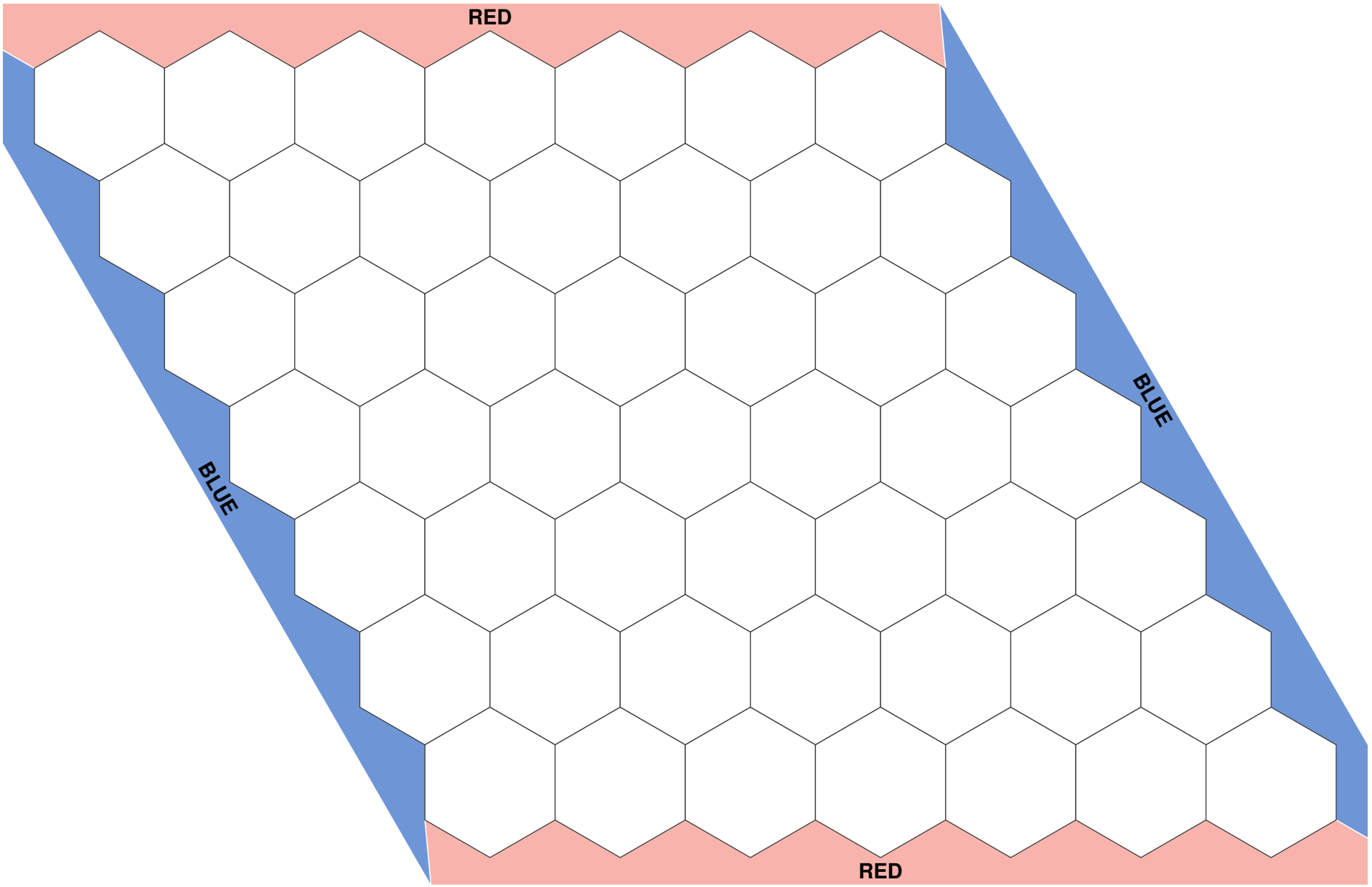




Problem 8: Explain why somebody always has a winning chain when every hexagon has a counter.

Problem 9: It is Red's turn in this game. Can Red make sure to win, however Blue plays? Next, put one more red counter on any empty hexagon. It is still Red's turn. Can Red still make sure to win? Can an extra red counter ever stop Red from winning?



Problem 10: Hex players often use a board with 11 rows of 11 hexagons. Nobody knows a sure way to win on it. Maya says she has a way to always win on that board when she goes second, playing by the rules on page 1. Explain why Maya must be wrong.

Problem 11: Play Hex on this board. Take turns going first.